



Checkpoint I: Project Proposal

Group: G08
Date: 2025/09/15

Problem Domain

Food is fundamental to human survival and has become central to numerous global industries. But is its distribution and production the same around the world? Has it remained constant throughout the years? How does it affect public health? These are some of the questions we ask ourselves when we watch the news or travel and notice the striking differences in people's lives across the globe.

This topic is especially **interesting** because diet is a **universal** human activity that everyone participates in differently, **constantly evolving** through culture, technology, and society while generating complex data that requires sophisticated tools to analyze and reveal meaningful patterns. It's also highly **relevant** since our food choices directly impact personal **health** and disease risk, drive **economic development** globally, help the food industry anticipate consumer needs, and increasingly influence both individual wellbeing and global **sustainability**, making clear and reliable dietary information more crucial than ever.

Task Abstraction

1. how daily calorie, protein, carbs and fat intake varies between countries?
 - **Abstract Form: How does W, X, Y and Z vary between different A?**
 - Goal: Compare how countries differ in their average intake of calories and macronutrients.
 - Visualization: A world map where each country is shaded based on nutrient intake levels.
 - Idiom choice: A world map is the clearest way to show country-level variation, since it makes geographic patterns easy to spot
 - Features:
 - Color hues represent consumption amount
 - Filters to switch between calories, carbs, protein, and fat.
 - Option to select and compare specific countries.
 - **Type: Geolocation**
 - Reasoning: Highlights cultural and dietary diversity across the world, reveals potential health risks.
2. How has the intake of calories, protein, carbs and fat changed throughout the years?
 - **Abstract Form: How does W, X, Y and Z change over A years?**
 - Goal: Explore historical trends in macronutrient consumption across countries.
 - Visualization: Line chart with intake trends over time.
 - Idiom choice: Line charts make it easy to see whether a nutrient's intake is rising, falling, or staying steady, and allow for clear comparisons.
 - Features:
 - X-axis: Years.
 - Y-axis: Amount of intake for the selected nutrient.
 - Filters to choose specific countries and nutrients.

- Option to compare multiple countries.
 - **Type: Evolution over time**
 - Reasoning: Shows long-term shifts in human nutrition, provides evidence for how urbanization, industrialization, and trade affect diets.
3. How much does each macronutrient contribute percentage wise to the total calories consumed in a day over different countries?
- **Abstract Form: How does each one of W, X and Y contribute to the total of Z consumed in A amount of time over different B?**
 - Goal: Show the proportion of daily calories coming from protein, carbohydrates, and fat.
 - Visualization: Pie chart of macronutrient distribution.
 - Idiom choice: A pie chart is well-suited for showing proportions and makes side-by-side comparisons between countries intuitive.
 - Features:
 - Country selector.
 - Percentage breakdown of calories by macronutrient.
 - Easy comparison of proportions between countries.
 - Type: Proportion
 - Reasoning: Exposes hidden imbalances in food intake
4. How does the top 10 countries in obesity vary along the years?
- **Abstract Form: How does the 10 biggest X in Y vary among Z years?**
 - Goal: Track how obesity rankings change across countries over time.
 - Visualization: Horizontal bar chart.
 - Idiom choice: A dynamic bar chart effectively illustrates ranking changes over time.
 - Features:
 - X-axis: Obesity rate
 - Y-axis: Top 10 ranked countries.
 - Bars reorder dynamically as years progress.
 - Type: Rank over time
 - Reasoning: Reveals how globalization and fast-food industry correlate with rising obesity
5. How does national wealth relate to intake?
- **Abstract Form: How does X relate to y?**
 - Goal: Investigate the relationship between economic development and calorie intake.
 - Visualization: Scatter plot comparing wealth and intake.
 - A scatter plot is ideal for showing correlations, it lets viewers see whether higher GDP per capita corresponds to higher calorie intake, while also highlighting outliers that don't follow the general trend.
 - Features:
 - X-axis: GDP per capita
 - Y-axis: Average daily calorie intake per person.
 - Option to hover/select countries for details.
 - type: Correlation
 - Reasoning: Raises questions about food accessibility, inequality, and sustainability, provides insights for global health organizations on how economic growth influences diets

Data Abstraction

Our first step was talking about common interests and the questions we wanted to answer. Then we started researching for datasets that had the required data.

All datasets set are static and available on these websites bellow, the data sets have 8 variables and +/- 35 000 items.

<https://ourworldindata.org/grapher/daily-caloric-supply-derived-from-carbohydrates-protein-andfat?country=USA~IND~PRT&overlay=download-data> <https://ourworldindata.org/grapher/share-of-adults-defined-as-obese?overlay=download-data> <https://ourworldindata.org/grapher/gdp-per-capita-maddison-project-database?overlay=download-data>

The data was already preprocessed by Our World in Data. They prepared the original data which involved several processing steps: standardizing country names and world region definitions, converting units, calculating derived indicators such as per capita measures, as well as adding or adapting metadata such as the name or the description given to an indicator.

For the first table we used the pre-prepared data, we had to clean useless columns (such as 3 letter codes representing country names already cited in other columns) and delete all the lines that contain information about regions ex: Africa because we only wanted countries. We also needed the percentage of all the macro nutrients in relation to the total calorie intake, so we had to calculate those derived measures as percentages of the total calorie intake.

For the second table we also had pre-prepared data we deleted useless columns (such as 3 letter codes representing country names already cited in other columns) and all the lines that contain information about regions for the same reason.

For the third and final table we also we used the pre-prepared data, we had to clean useless columns (such as 3 letter codes representing country names already cited in other columns) and delete all the lines that contain information about regions, then we calculated the total amount of calories using the first original dataset and then we did an inner join with the GDP data using Country and Year as joining attributes (The calories table had information from 1960 to 2022, while the GDP table had data ranging from 1900 to 2022, in order to not have a big number of items without 90% of the columns, the items with years before 1960 were dropped).

Description of the final dataset

Dataset type: table

Attribute Name	Type	Features	Semantics
Years	Continuous	Interval	Represents the year when data was recorded
Carbohydrates	Ratio	Absolute	Daily calorie supply per person from carbohydrates
Fat	Ratio	Absolute	Daily calorie supply per person from fat
Entity	Nominal	—	The country where the data was collected.
GDP per capita	Ratio	Absolute	A country-level measure of economic status
Obesity	Ratio	Percentage	Percentage of the population classified as obese
Animal protein	Ratio	Absolute	calorie supply per person that comes from animal protein,

Vegetable protein	Ratio	Absolute	Daily calorie supply per person that comes from vegetal protein
Total calories	Ratio	Absolute	Total daily calorie supply per person
Animal protein percentage	Ratio	Percentage	Percentage of the total calories that comes from animal protein
Vegetable protein percentage	Ratio	Percentage	Percentage of the total calories that comes from vegetable protein
Fat percentage	Ratio	Percentage	Percentage of the total calories that comes from fat
Carbohydrates percentage	Ratio	Percentage	Percentage of the total calories that comes from carbohydrates

Mapping

1. We aim to understand how daily intake of calories, Animal protein, Vegetal protein, carbohydrates and fat varies across countries. By analyzing the corresponding attributes daily calorie intake, Animal protein, Vegetal protein, carbohydrates and fat Animal protein, from the table `Total_calorie_distribution_per_macronutrient` and linking them to their respective countries, we can explore how these nutritional factors are distributed geographically.
2. We want to visualize how daily intake of calories, Animal protein, Vegetal protein, carbohydrates, and fat has changed across countries over time. By linking the attributes daily calorie, Animal protein, Vegetable protein, carbohydrate, and fat intake from the table `Total_calorie_distribution_per_macronutrient` to their respective countries and comparing them over time periods on the same table, we can compare trends and observe how the distribution of these dietary components has evolved within each country.
3. We want to study How much does each macronutrient contribute percentage wise to the total calories consumed in a day over different countries. To do this, we will use a derived measure that converts the intake of specific macronutrients into their caloric equivalents, based on the conversion factors provided in Table `Total_calorie_distribution_per_macronutrient`. Then, using these values, we will calculate the proportion each macronutrient contributes to the total daily caloric intake.
4. We aim to analyze how the top 10 countries by obesity rate have changed over the years. By using the obesity rate data from table `Obesity_rate_per_year` and linking it to the corresponding countries and years from the same table we can generate ranked lists for each year and visualize the shift in rankings across time to identify global trends in obesity.
5. We want to explore the relationship between national wealth and dietary intake. By linking GDP per capita from table `Daily_calories_X_GDP_per_capita` with the daily calorie intake values from the same table , we can construct scatter plots that reveal potential correlations between wealth and nutritional consumption across countries. This analysis will also highlight outlier countries those with unusually high or low intake relative to their economic status helping us investigate exceptional cases that deviate from overall trends.